

Yarn Architecture

Need of yarn

Hadoop 1.x vs Hadoop 2.x

YARN architecture

Resource Manager

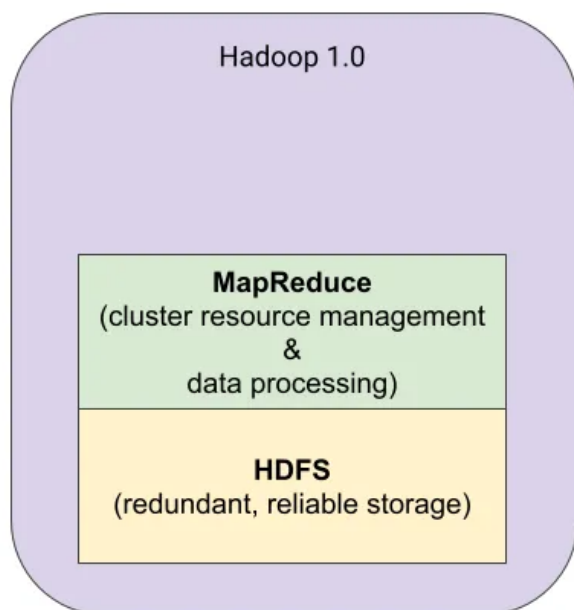
Node Manager

App Master

Container

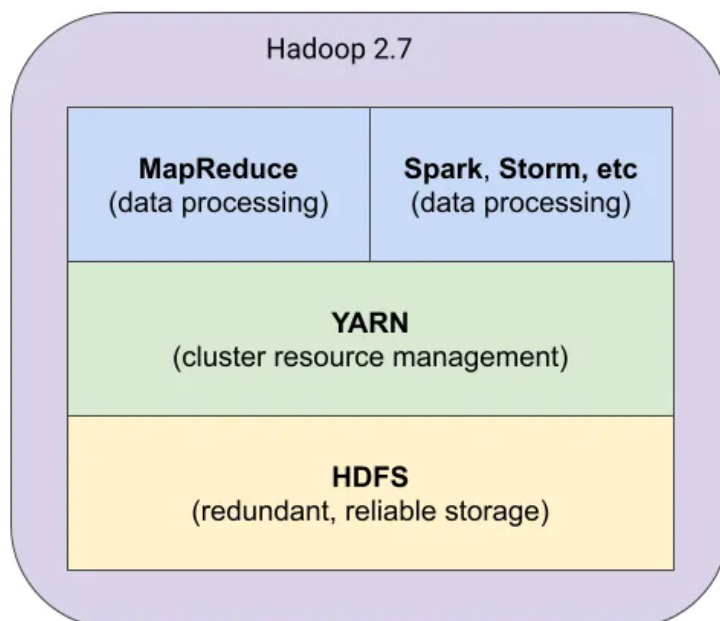
Before 2012

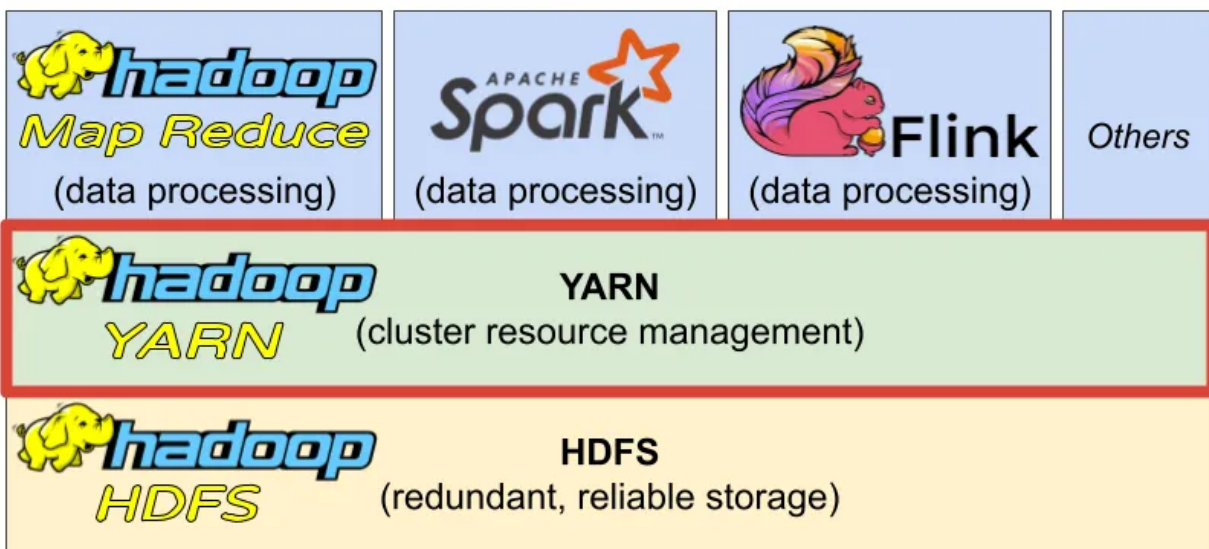
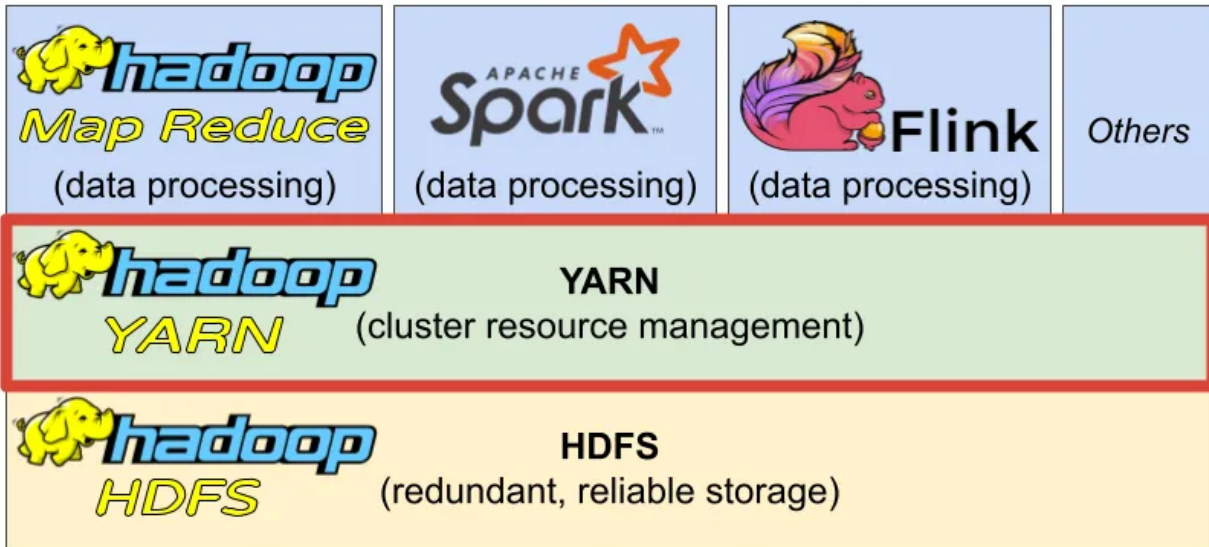
Users could write MapReduce programs using scripting languages

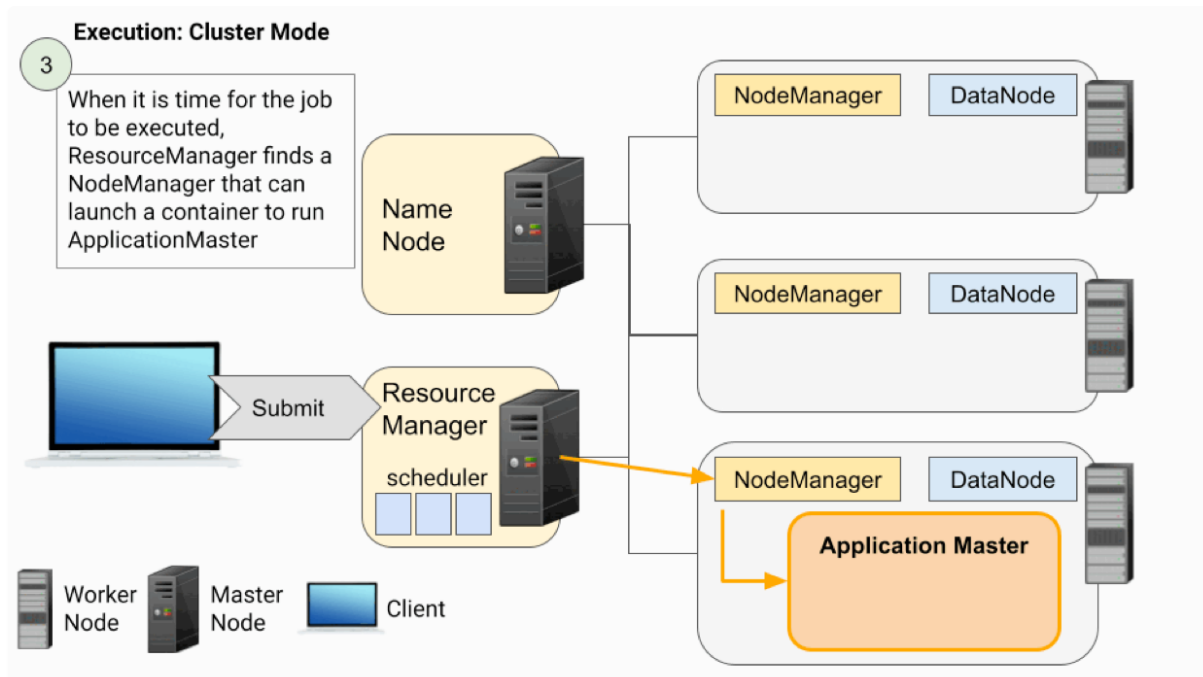


Since 2012

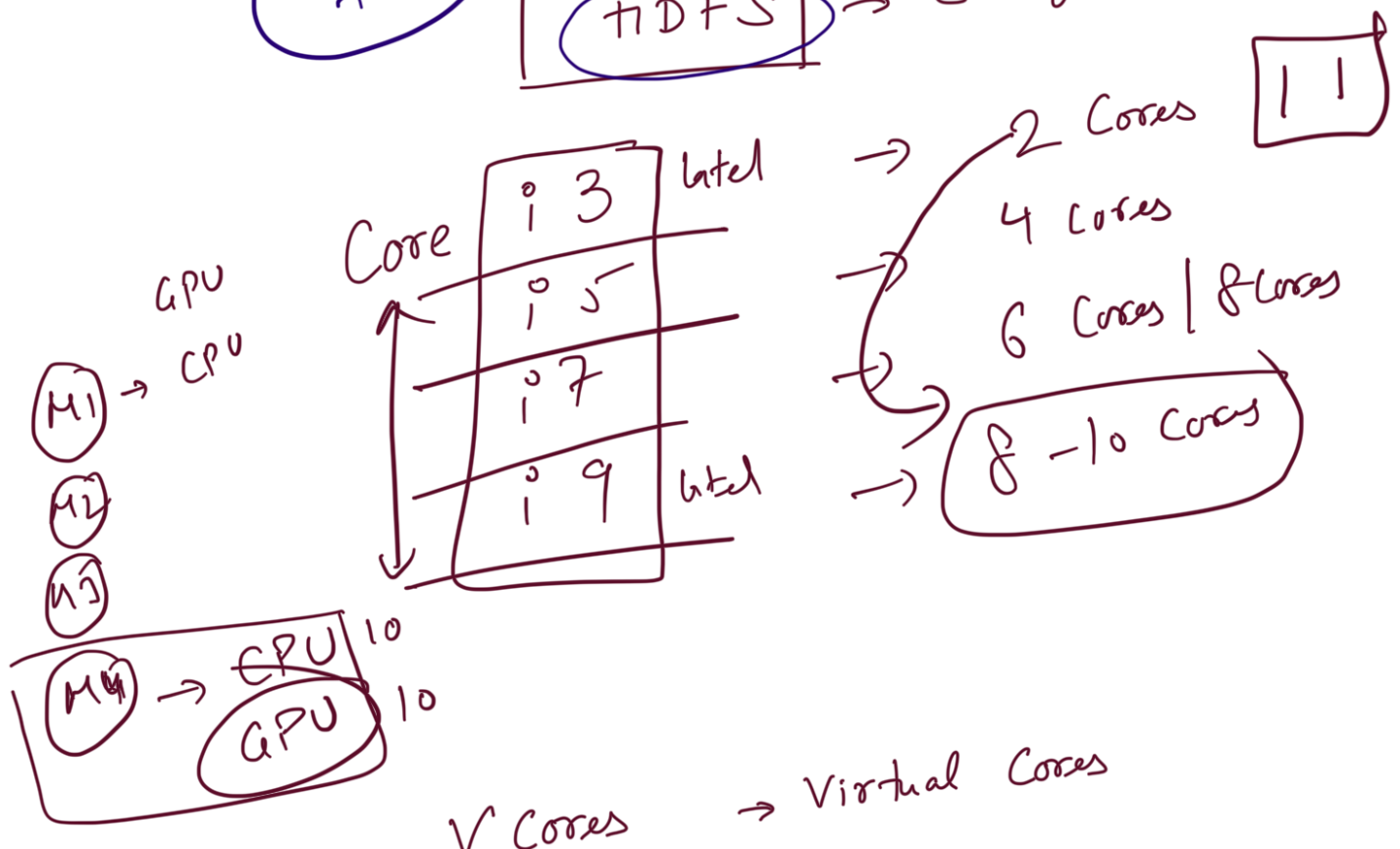
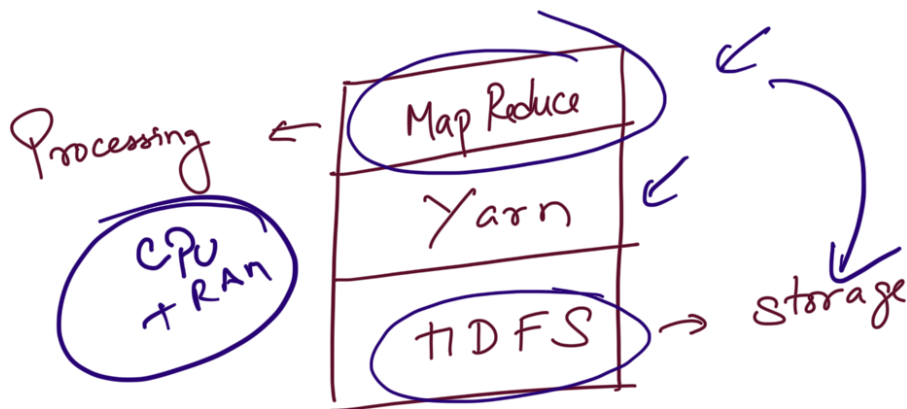
Users could work on multiple processing models in addition to MapReduce



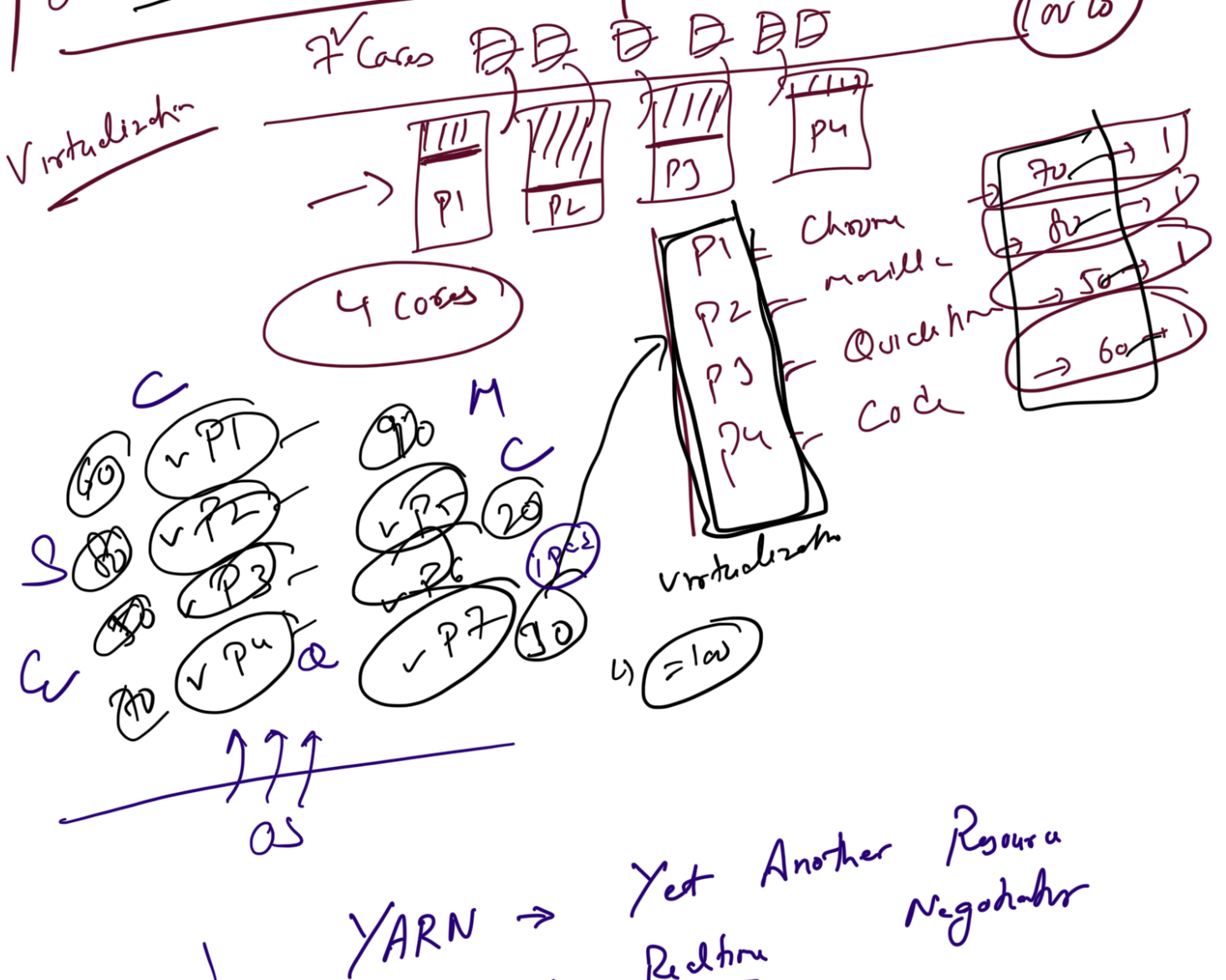


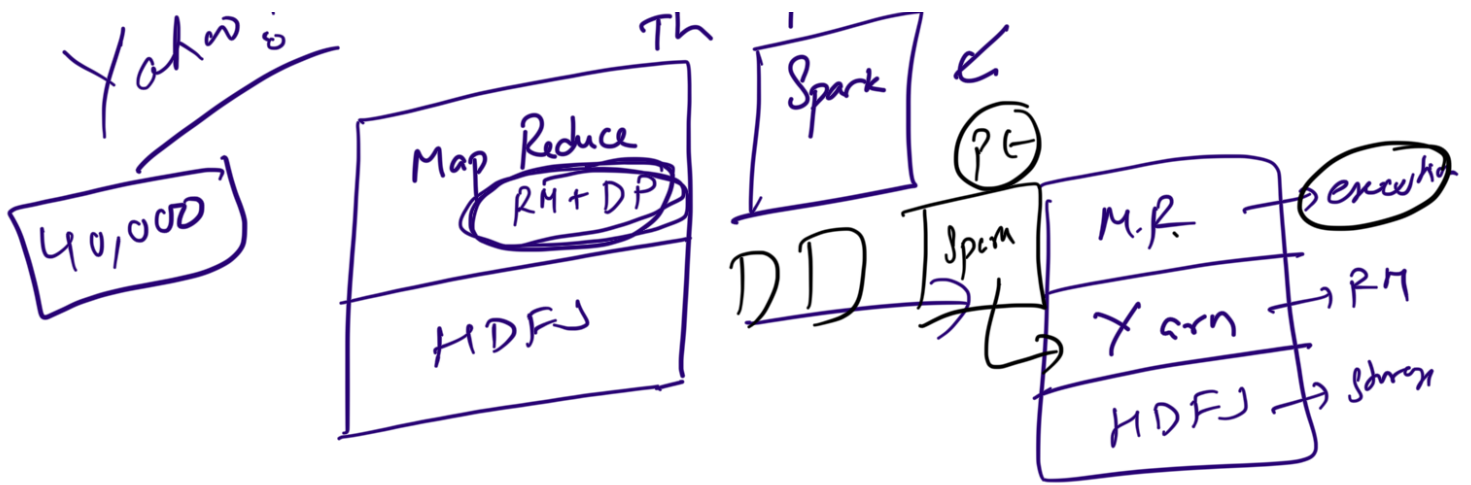


Yarn

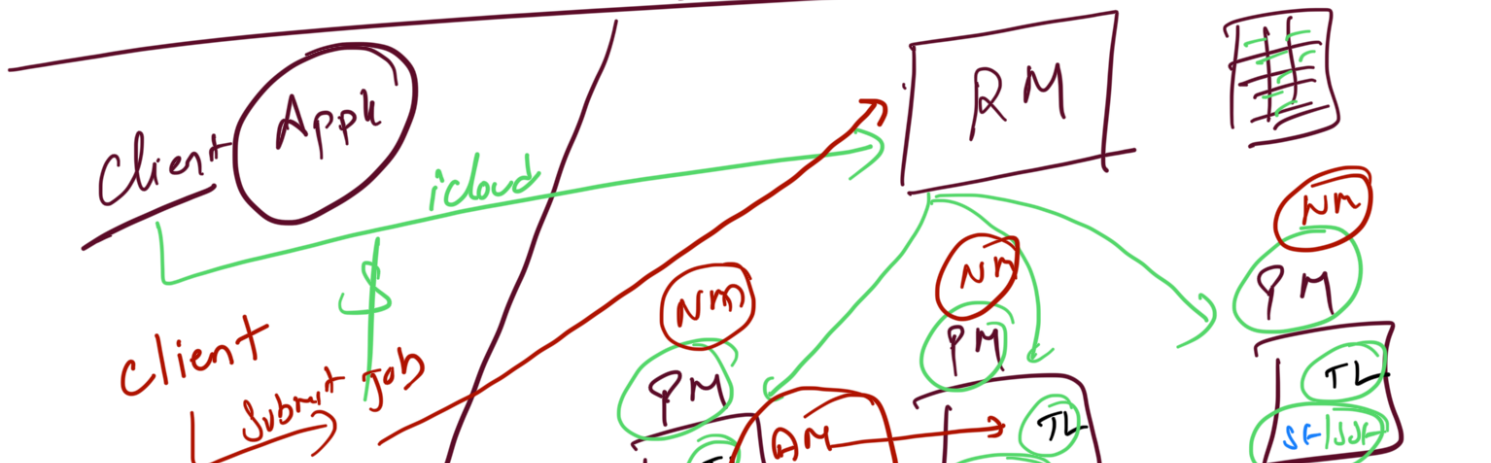
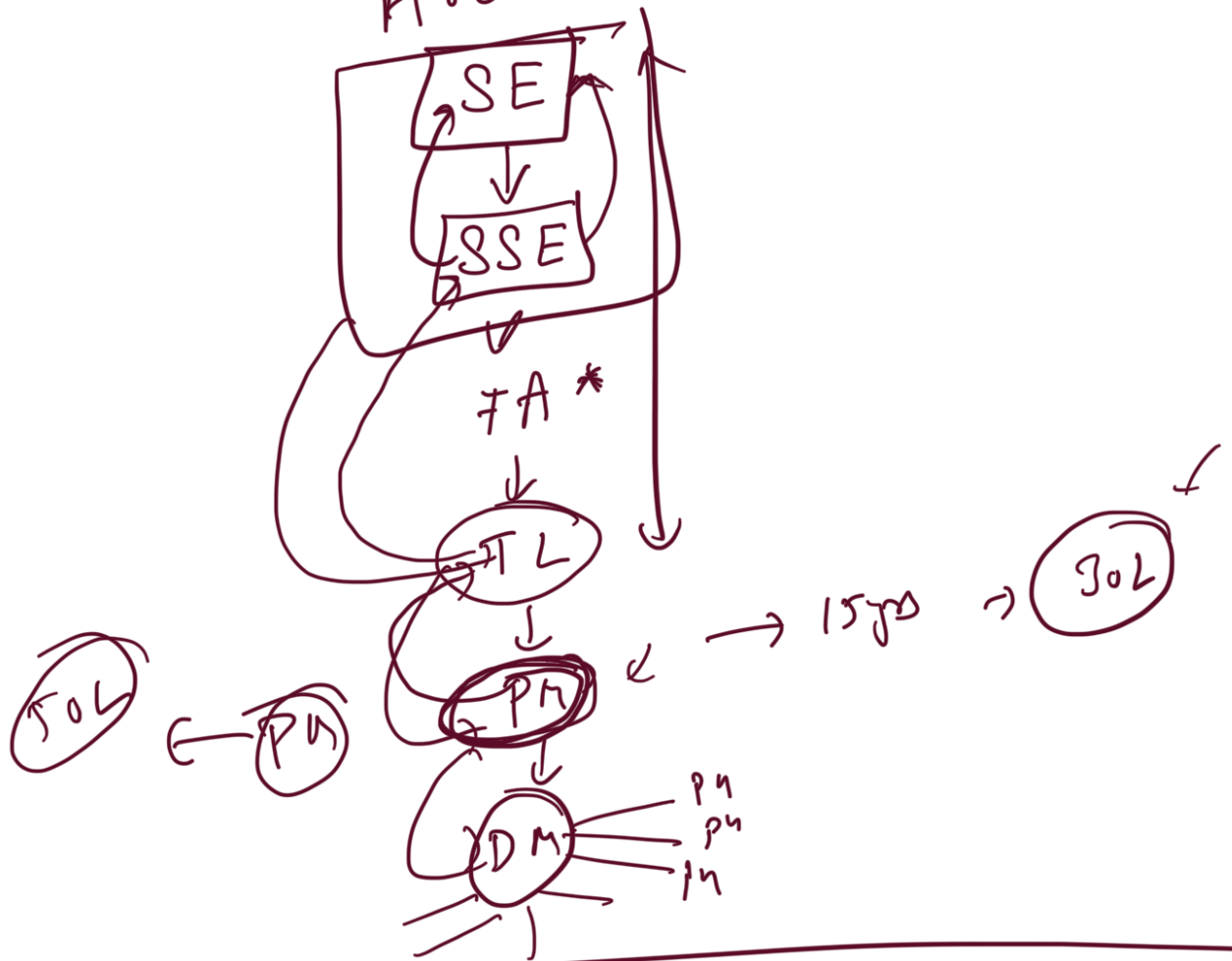


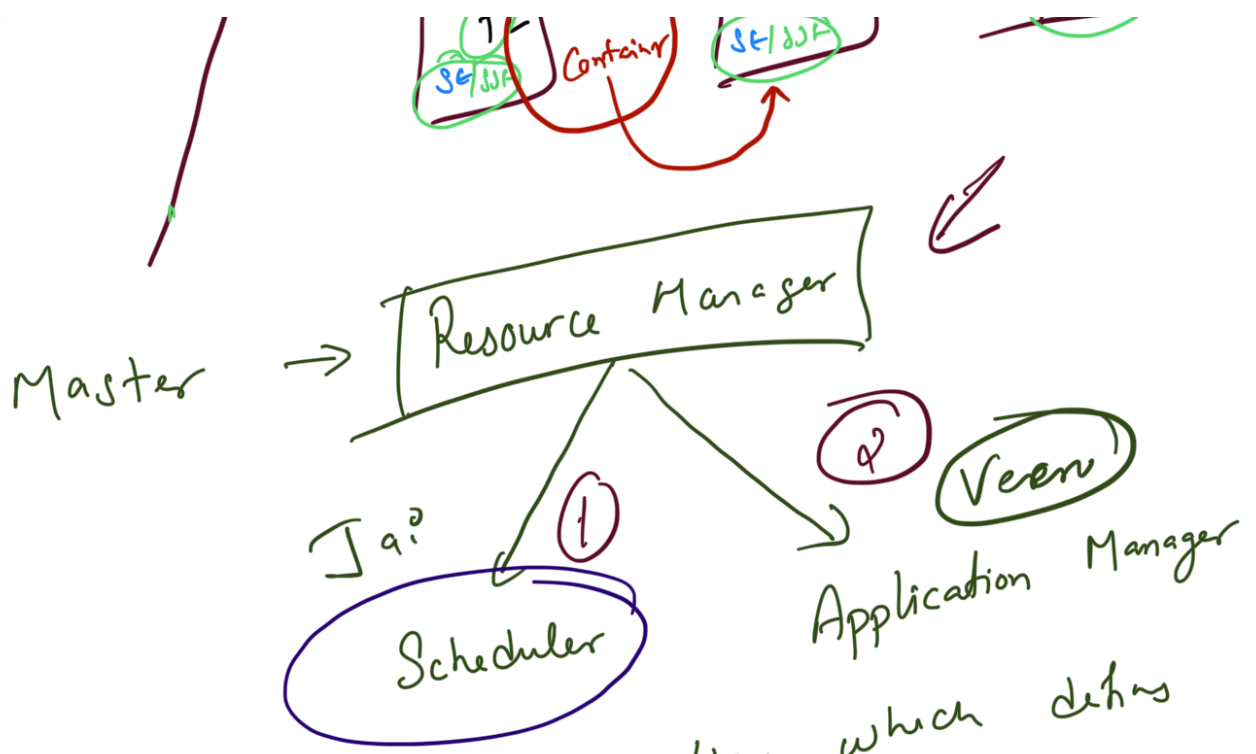
Core	V Cores
→ represents a <u>Physical</u> CPU is a cluster → at OS level has Computers 4 Cores ↓ 2 0.6 Cores = 1 V Cores 7 V Cores	→ represents a <u>logical</u> CPU unit. → 1st (cluster) level for scheduling workloads. 40% (7) 0.6 x 8 (100%)





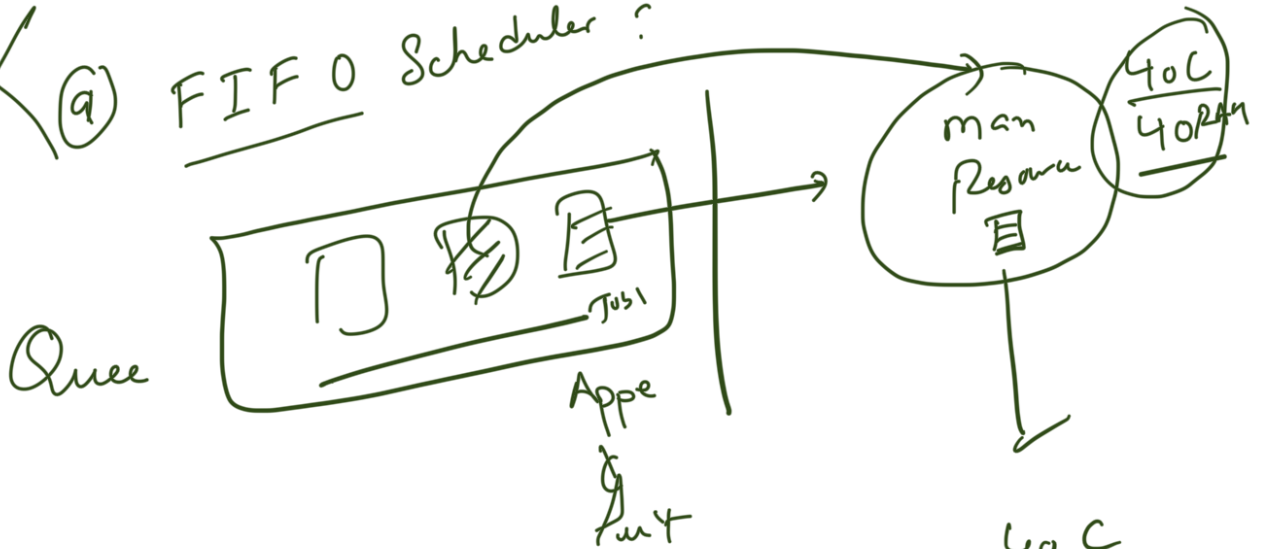
Architecture





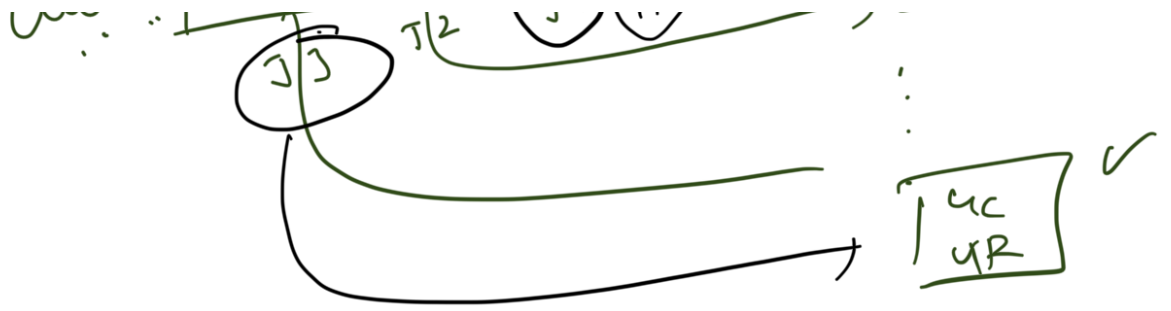
① Scheduler :- has a policy which defines how a job should be executed :

X (a) FIFO Scheduler ?



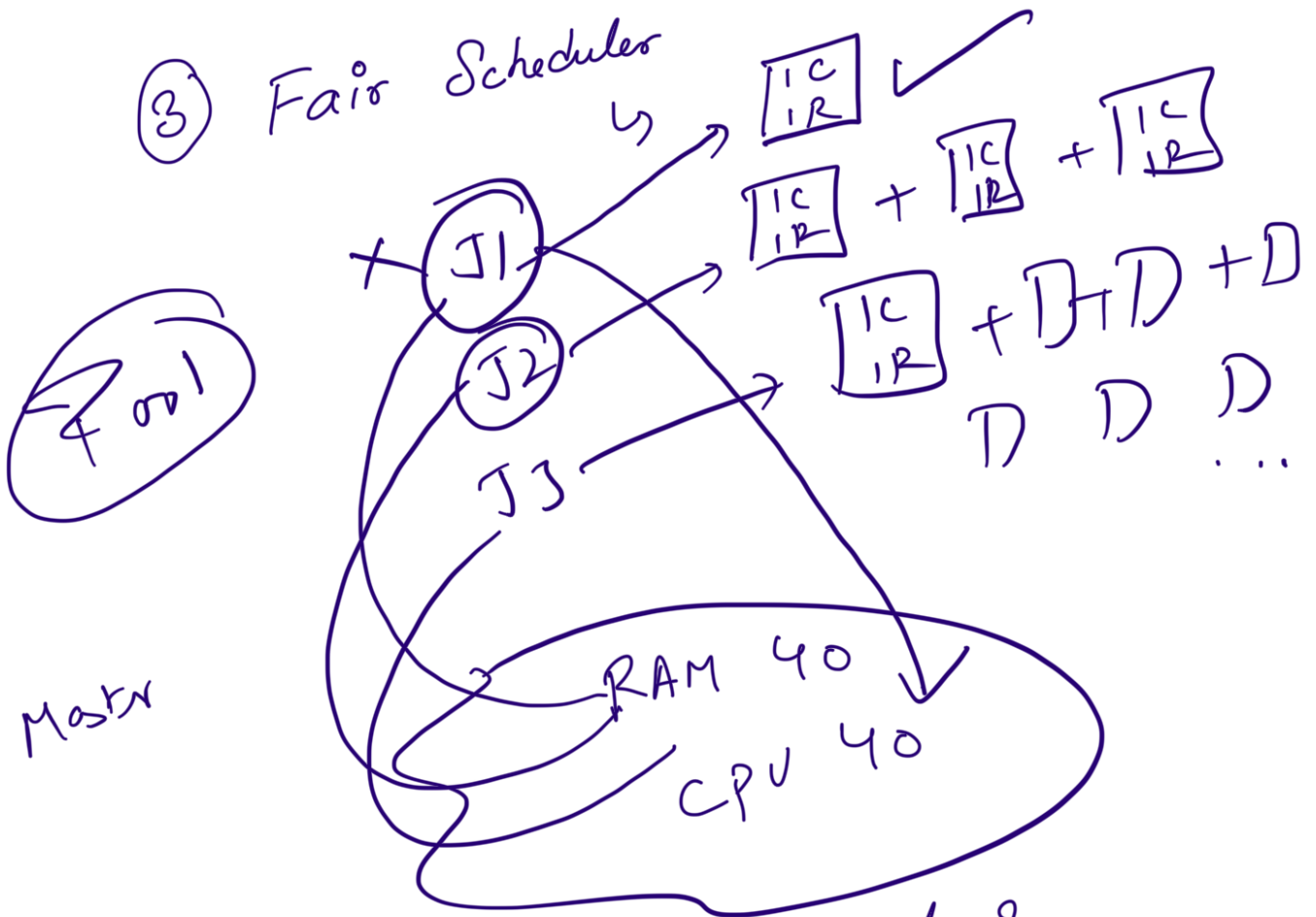
(b) Capacity Scheduling





40%	→ (DS) team
30%	→ DA
30%	→ Real time Analyhy

③ Fair Scheduler



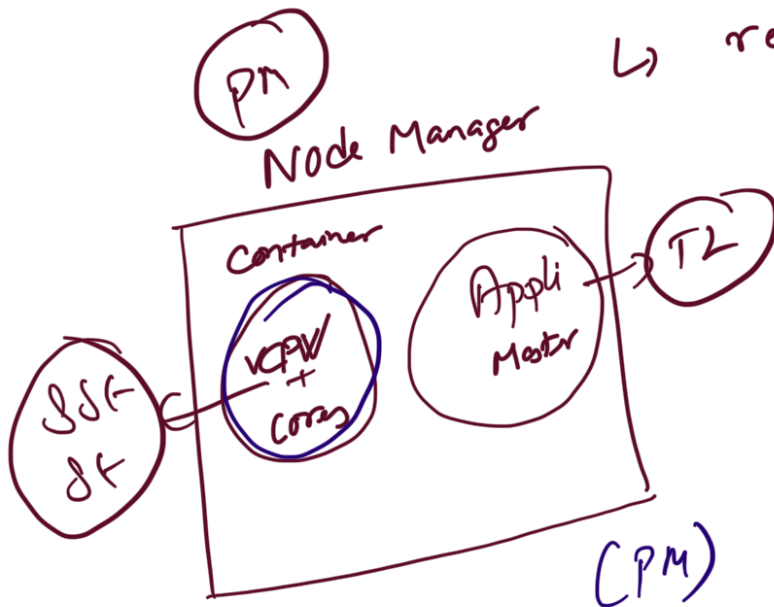
what Scheduler is not meant to do?

↳ restart the failed tasks.
not monitor or track the

↳ does not states of application.

Application Manager :-

↳ accepting job Submission
is negotiate resources
↳ restart the job on failure.



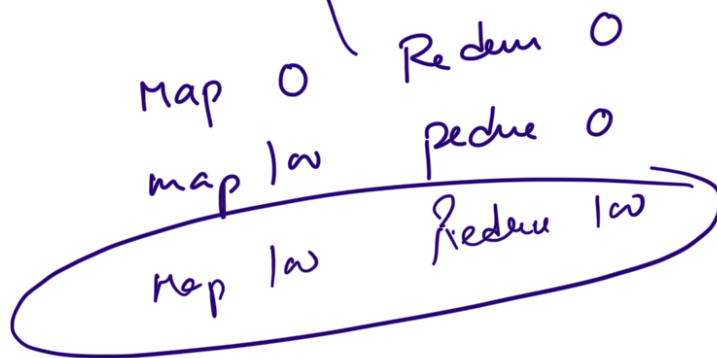
① Node Manager :-

↳ Container Dependencies
↳ Node health
↳ Reporting Container status to RM.
↳ Container Lifecycle mgmt.

② Application Master (jar) :-

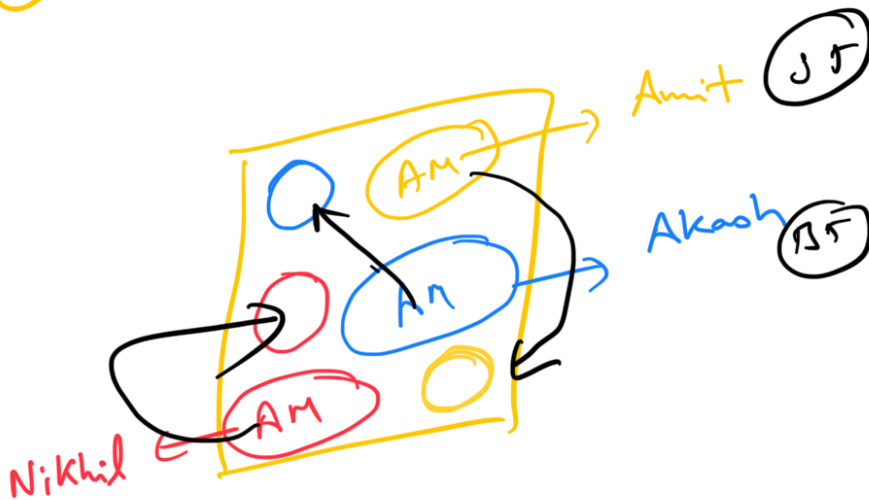
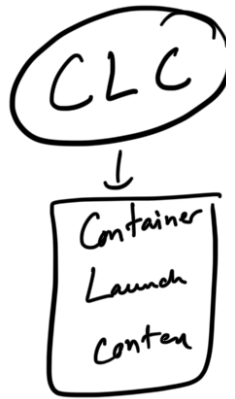
↳ managing Application lifecycle.
↳ dynamic adjustments to resource consumption.
→ manage faults & metrics

→ provides status & metrics to: NM, RM.



999
109Ks

③ Set/Get Containers :- RAM + vCPU
→ Security tokens



Step 1 :- The Application Master Contacts the related NM.

Step 2 :- The Container executes the Appl Master

Step 3 :- The RM allocates a container
the client submits an appl to RM.

Step⁴ :

Step⁵ : the NM launches the Contain...

4 3 1 5 2